

MediKiosk AI Architecture vs. MedGemma / Med-Gemini

A Comparative Evaluation for Multilingual Outpatient Triage & Prescription Digitization

Executive Summary: MediKiosk employs a hybrid multi-tiered architecture (Google Gemini Flash + Groq Llama-3.1-70B + Ollama Qwen-2.5 + Sarvam AI + Multilingual Sentence Transformers) tailored for high-speed, low-latency, Indic vernacular triage and edge resilience in Indian clinics. In contrast, **Google's MedGemma & Med-Gemini** represent domain-specialized medical foundation models tuned for clinical examinations (USMLE), EHR summarization, and radiology imaging. This report analyzes why MediKiosk's hybrid approach meets kiosk field requirements while identifying where MedGemma models provide clinical advantages.

1. Architecture & Model Overview

Evaluation Axis	MediKiosk Multi-Tier Stack	Google MedGemma / Med-Gemini
Core Models	Tier 1 (Cloud Vision/Quality): Gemini 2.5 Flash Tier 2 (Cloud Speed): Groq Llama-3.1-70B Tier 3 (Edge/Offline): Ollama Qwen-2.5 7B/3B Speech/Indic: Sarvam AI + IndicConformer	MedGemma: 2B, 9B, 27B open medical LLMs Med-Gemini: Multimodal clinical foundation models (Gemini-based with medical RLHF/tuning)
Domain Tuning	General foundation models reinforced with structured JSON schemas , concept bank matching, and drug-interaction engines.	Pre-trained & fine-tuned on biomedical benchmarks (PubMed, MedQA/USMLE, MIMIC, clinical EHR notes).
Deployment Topology	3-Tier Hybrid Failover: Cloud Quality & Speed with instant on-premise local GPU/CPU fallback.	MedGemma: On-premise server (requires 16–32GB+ VRAM). Med-Gemini: Cloud API (Vertex AI Healthcare).

2. Deep-Dive Comparative Analysis

A. Multilingual & Indic Dialect Processing (Crucial for Indian Kiosks)

- MediKiosk Stack:** Combines Gemini Flash and Qwen-2.5 (strong Indic tokenization) with **Sarvam AI** and sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2. Effectively handles Hindi, Marathi, Tamil, Telugu, and Hinglish code-switching (e.g., patient colloquial complaints like '*chhati me jalan*' or '*chakkar aana*') and maps them directly to standardized SNOMED-CT / NAMASTE (AYUSH) clinical concepts.
- MedGemma / Med-Gemini:** MedGemma is primarily tuned on English biomedical literature and US clinical terminology. Its comprehension of regional Indic scripts and colloquial Indian symptom expressions is limited without custom fine-tuning.

B. Prescription OCR & Multimodal Clinical Vision

- MediKiosk Stack:** Leverages Gemini Flash Vision for zero-shot extraction of handwritten Indian doctor prescriptions, dosages, frequencies, and lab report tables, with automatic fallback to local RapidOCR + Drug Interaction Engine.
- Med-Gemini:** Industry state-of-the-art on complex radiological imaging (Chest X-Rays, CT scans, Pathology) and multi-page EHR records. However, base MedGemma text models lack integrated vision capabilities and require separate vision pipelines.

C. Latency, Hardware Footprint & Operational Cost

- **MediKiosk Stack:** Groq Llama-3.1-70B achieves sub-500ms TTFT (Time To First Token) for real-time natural dialogue. For offline kiosks, Qwen-2.5 3B/7B runs efficiently on modest hardware (8GB–16GB VRAM or quantized CPU). Free tier quotas (Gemini 1,500 req/day, Groq 14,400 req/day) allow zero API costs for pilot kiosk deployments.
- **MedGemma:** MedGemma 9B / 27B requires dedicated GPU compute (16GB–32GB+ VRAM) at each kiosk node or high server hosting costs, making distributed low-cost rural kiosk hardware significantly more expensive.

D. Clinical Safety, Diagnostic Depth & Hallucination Guardrails

- **MediKiosk Stack:** Enforces strict Pydantic JSON schemas, deterministic drug-drug interaction matrix checks, and concept bank matching (concept_bank.json) to prevent general LLM hallucinations during triage summary generation.
- **MedGemma / Med-Gemini:** Features native biomedical reinforcement learning (RLHF with medical doctors), leading to superior diagnostic reasoning on complex differentials, rare pathology, and medical board benchmarks.

3. Capability & Feasibility Scorecard

Evaluation Criteria	MediKiosk Stack	MedGemma / Med-Gemini	Optimal Application
Indic Vernacular & Code-Mixing	Superior (Sarvam + Multilingual)	Moderate (English-centric)	Indian Primary Care & Kiosks
Doctor Prescription Vision	High (Gemini Flash Vision)	SOTA (Med-Gemini for Scans)	Prescription Digitization / Radiology
Turn-Taking Latency (<500ms)	Ultra-Fast (Groq LPU Engine)	Standard (Compute bound)	Interactive Patient Voice Triage
Low-Cost Edge / Offline Feasibility	Excellent (Qwen 3B/7B low VRAM)	Demanding (16GB+ VRAM required)	Rural Kiosks with Intermittent Internet
Deep Clinical Reasoning / USMLE	Prompt-Guided	Clinically Fine-Tuned	Specialist Diagnostic Support

4. Strategic Conclusion & Recommendations

Key Takeaway: For MediKiosk's operational requirements — an interactive, multilingual triage kiosk deployed in Indian outpatient settings — **the current multi-tiered hybrid stack is the optimal architecture.** It provides unmatched conversational speed (Groq), superior Indic dialect recognition and ontology mapping (Sarvam + Qwen + Paraphrase-MiniLM), and robust zero-internet edge survivability.

Future Integration Pathway: If MediKiosk expands into specialized secondary hospital workflows (e.g., automated differential diagnosis generation, Chest X-ray triage, or complex EHR multi-specialist consultations), integrating **MedGemma as a dedicated clinical second-opinion service** in the cloud tier would provide substantial value.